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Effects of endodontic instrument handle diameter on electromyographic activity of forearm and hand muscles

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Abstract

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Aim To determine the influence of the handle diameter of endodontic instruments on forearm and hand muscle activity using electromyographic (EMG) recording.

Methodology Size 45 K-type files were fitted with four different handle diameters; 3.5, 4.0, 5.0, and 6.0 mm. Seven dentists then attempted to negotiate to the working length acrylic resin root canals with each of the four handle sizes using a reaming motion. EMG activities were recorded from the flexor pollicis brevis muscle (f.p.b.), the flexor carpi radialis muscle (f.c.r.), and the brachioradialis muscle (b) with bipolar surface electrodes. The time taken to negotiate the canals, the area of integrated EMG that corresponded to the amount of EMG activity required during penetration and the maximum amplitude of EMG were measured using the EMG data. Results were analysed statistically using a one-way factorial ANOVA test and multiple comparison tests.

Results Reaming time and integrated EMG area of each muscle decreased with an increase in handle diameter. The most significant difference in time and area of integrated EMG was detected between handles of 6 mm and 3.5 mm diameter (time: $P < 0.01$, area of the f.p.b.: $P < 0.01$, area of the f.c.r. and b: $P < 0.05$), and between handles of 5 mm and 3.5 mm diameter ($P < 0.05$). Both 5 mm and 6 mm handles significantly decreased the maximum amplitude of EMG recorded from the f.p.b. compared with 3.5 mm handles (between 3.5 mm and 6 mm: $P < 0.01$, between 3.5 mm and 5 mm: $P < 0.05$).

Conclusion The results indicate that handle diameter has an effect on reaming time as well as on muscle activity. As a consequence, handle diameter influenced operator performance during instrumentation.

Keywords: EMG, endodontic instrument, forearm, hand, handle diameters.

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Introduction

As root canal preparation by hand instruments requires substantial effort, operators often suffer pain or fatigue in their arms and hands after extended treatment duration. A survey of 192 dentists described which symptoms were experienced in the hands and forearms of dentists during their dental practice (Ozawa *et al.* 1998). The results indicated that 79.2% had experienced symptoms in their hands and forearms, and the symptoms were highest during or after root canal preparation.

Although many mechanical root canal preparation systems (Abou-Rass & Jastrab 1982, Siamos *et al.* 1987, Kobayashi *et al.* 1998) have been developed to increase treatment efficiency, many practising dentists still use hand instruments because of their tactile feedback and fine digital control. In order to reduce operator effort, the design of instruments and instrumentation techniques requires further consideration.

Whilst the handle design of commercial endodontic

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instruments are barrel-type, round-type, or concave-type, most handle diameters were in the 3.5~4 mm range. When a sleeve is attached to a handle the handle diameter may become more than 6 mm (Wallen & Chandler 1998).

Chandler & Bloxham (1990) and Treble *et al.* (1993) discussed commercially available handles with different designs and diameters. Chandler *et al.* (1996) found no significant difference in operator performance during root canal instrumentation between handles of identical design but with four diameters 2.5~5 mm. The experiment only focused on performance time and did not look at operator effort.

Electromyographic recording has been used as a means of investigating muscle activity in the fields of medicine and biology since Adrian & Bronk (1929) reported the physiological significance of EMG recorded from the skeletal muscle. In dentistry, Mayers (1949) introduced EMG of temporomandibular muscle contraction patterns; subsequently Kawamura & Majima (1964), Ahlgran & Owall (1970) and Kohyama *et al.* (1998) analysed mainly the muscles of mastication using EMG.

Most of the symptoms experienced in the hands and forearms of dentists during or after root canal preparation were principally fatigue or pain (Ozawa *et al.* 1998). These symptoms might occur from excessive muscle activity.

Hand instrument motions can be classified into reaming or filing; a reaming action needs more effort as its purpose is to penetrate the root canal. A reaming action involves holding the handles of instruments with the pulp of the thumb and index finger and rotating. This is not a normal activity and there are no previous reports about the relation between reaming action and muscle activity.

This experiment completed using electromyographic (EMG) recording aimed to determine whether the handle diameter of endodontic instruments was an important factor in the efforts expended by dentists.

Materials and methods

Size 45 K-type files were fitted with four different diameter handles, namely 3.5, 4.0, 5.0, and 6.0 mm (PEACE Co., Tokyo, Japan) (Fig. 1). All handles had identical round designs and the same vertical ribbings; the instrument length was always 25 mm. Acrylic resin root canal models (working length: 5 mm, inner diameter: 0.5 mm) were also constructed. The subjects were seven practising dentists with less than 1 year clinical experience (male, 24~28-year-old).

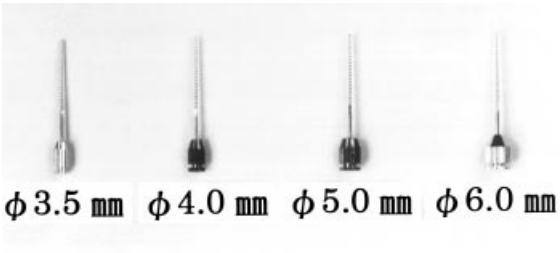

Figure 1 K-type files with different handle diameters.

The dentists sat in their normal operating position and held the handle using their thumb and index finger whilst wearing well-fitting latex finger guards. They were instructed not to touch the canal model with their fingers and to rotate the handle carefully in order to avoid adverse events, yet achieve the goal as quickly as possible. After three practices with each of the different handle diameters, the dentists then endeavoured to penetrate the canal models with instruments having one or other of the four handle sizes; the sequences of use of the different handles were randomized. All used a reaming motion until reaching the working length. If an instrument fractured during penetration and blocked the canal, the specimen was withdrawn from the analysis; the numbers of fractured files and canal blockages were recorded. In these cases, the subject used another file and a fresh canal model for the time/effort analysis.

Three muscles that exhibit a strong movement during the holding/pinching motion on palpation (Martini & Timmons 1995, Satoh *et al.* 1969) were identified (Fig. 2):

- the flexor pollicis brevis muscle (f.p.b.)
- the flexor carpi radialis muscle (f.c.r.) and
- the brachioradialis muscle (b).

The EMG activities of these three muscles were recorded using bipolar surface electrodes (Fig. 3). Prior to taking the EMG recordings, the EMG activity of the three muscles was compared to confirm whether the EMG was being recorded from each muscle. Subjects were instructed to do three tasks (thumb–index finger pulp pinch, dorsiflexion, palmar flexion) as shown in Fig. 4.

The EMG activities recorded from the three muscles during the reaming motion were amplified (BIOELECTRI AMPL. 1253 A; NEC Medical Systems, Tokyo, Japan) and integrated (INTEGRATOR 1322; NEC Medical Systems, Tokyo, Japan) with a time constant of 0.1 s. The EMG and integrated EMGs were recorded simultaneously

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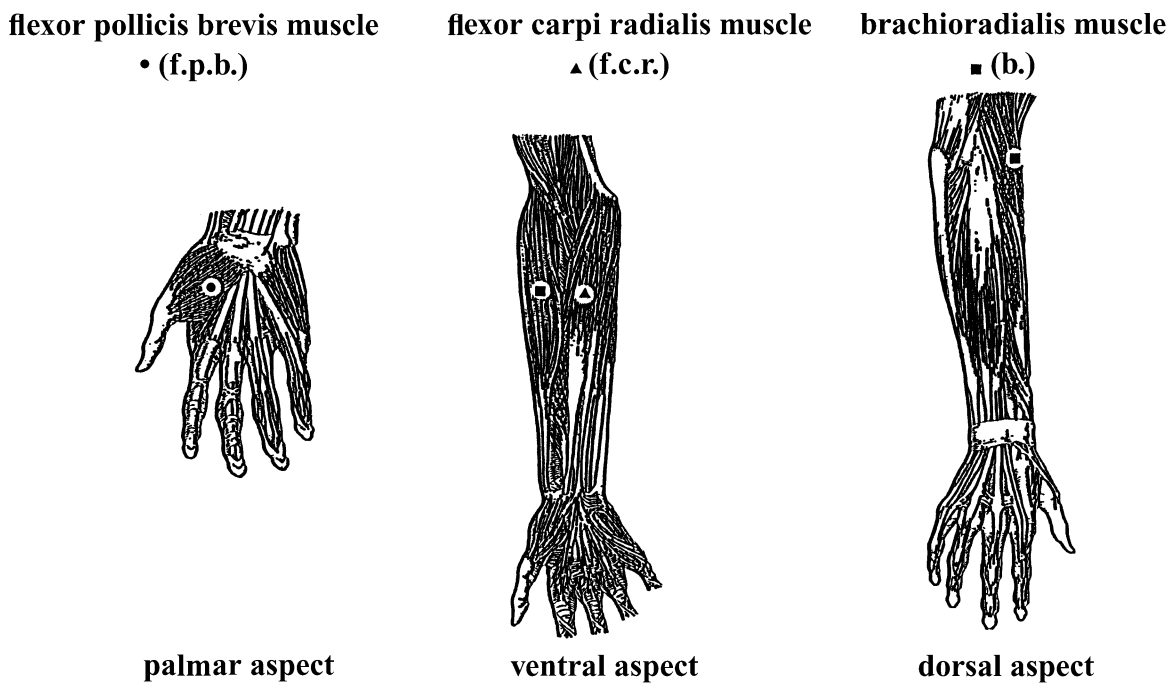

Figure 2 Muscles used during evaluation.

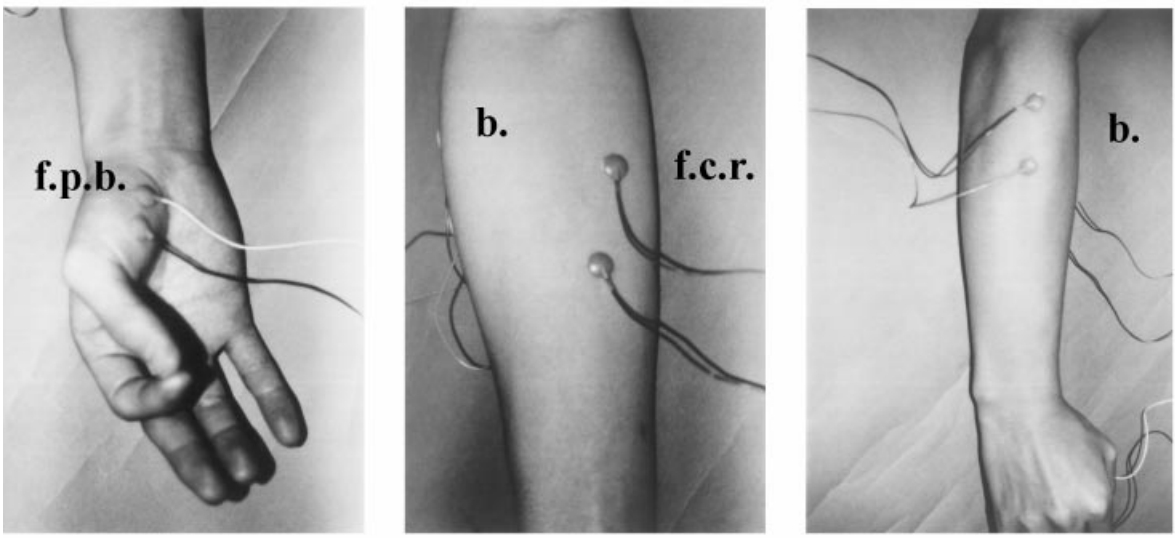

Figure 3 Bipolar surface electrodes placed on the tested muscles.

on a pen-recorder (RECTI–HORIZ 8K; NEC Medical Systems, Tokyo, Japan) for further analysis. A metal plate connected to a pen-recorder was placed under the root canal model. When the tip of a file reached the working length and connected with the plate, an electric signal (3 V) was generated, and this signal of penetration was recorded on a pen-recorder along with the EMGs.

Using the recorded EMG data, the reaming time, the integrated EMG area which corresponded to the

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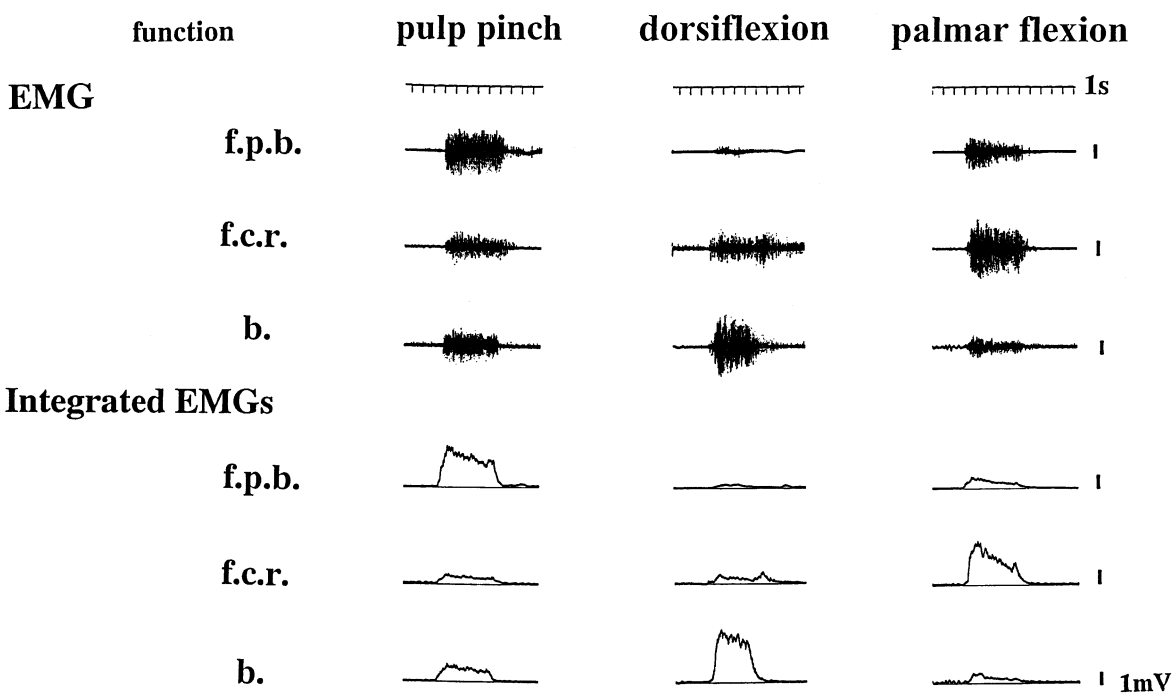

Figure 4 EMG activities of three muscles during pulp pinch, dorsiflexion and palmar flexion.

amount of EMG activity required for penetration, and the maximum amplitude of EMG for reaming motion to penetration were graphically rendered on a personal computer with appropriate software (PhotoShop ver.4.0, Adobe Systems Inc., San Jose, CA, USA), and NIH Image ver.1.56 (National Institute of Health, WA, USA).

Statistical means were calculated for the four experimental groups, and a one-way factorial ANOVA test was used to compare the handle sizes. Following the ANOVA test, all paired differences between means were tested using multiple comparison tests (Fisher's Protected Least Significant Difference).

Results

Typical records of EMGs during reaming motion are shown in Fig. 5. The results are summarized in Table 1.

The 3.5 mm handle diameter file required the longest instrumentation time, and this proved to be significantly longer than the time required for either the 5 mm ($P < 0.05$) or 6 mm ($P < 0.01$) handles. Comparison of 3.5 mm and 4 mm handles showed that there were no significant differences; whereas the best times were recorded with a 6-mm handle, this did not prove to be significantly faster than either the 4 mm or 5 mm handles. The integrated EMG area of each muscle decreased with increasing handle diameter. The integrated EMG area from each muscle with both 5 mm and 6 mm handles significantly decreased when compared with the 3.5 mm handle (only the f.p.b. between 3.5 mm and 6 mm: $P < 0.01$, the others: $P < 0.05$). There was no significant difference in the integrated EMG area between 5 mm and 6 mm handles or between 3.5 mm and 4 mm handles.

The maximum amplitude of EMG recorded from the f.p.b. decreased following an increase in handle diameter, and both 5 mm and 6 mm handles significantly decreased the maximum amplitude of EMG from f.p.b. compared with the 3.5 mm handle (between 3.5 mm and 6 mm: $P < 0.01$, between 3.5 mm and 5 mm: $P < 0.05$). A significant difference in the maximum amplitude of EMG from the f.p.b. was noted between 4 mm and 6 mm handles ($P < 0.05$), but not between 3.5 mm and 4 mm. A comparison of handle sizes yielded no significant differences in the maximum amplitude of EMG recorded from the f.c.r. and the b. On comparison of 4 mm and 6 mm handle diameters, there were significant differences exhibited in the integrated EMG area and the maximum amplitude of EMG from only the f.p.b. ($P < 0.05$).

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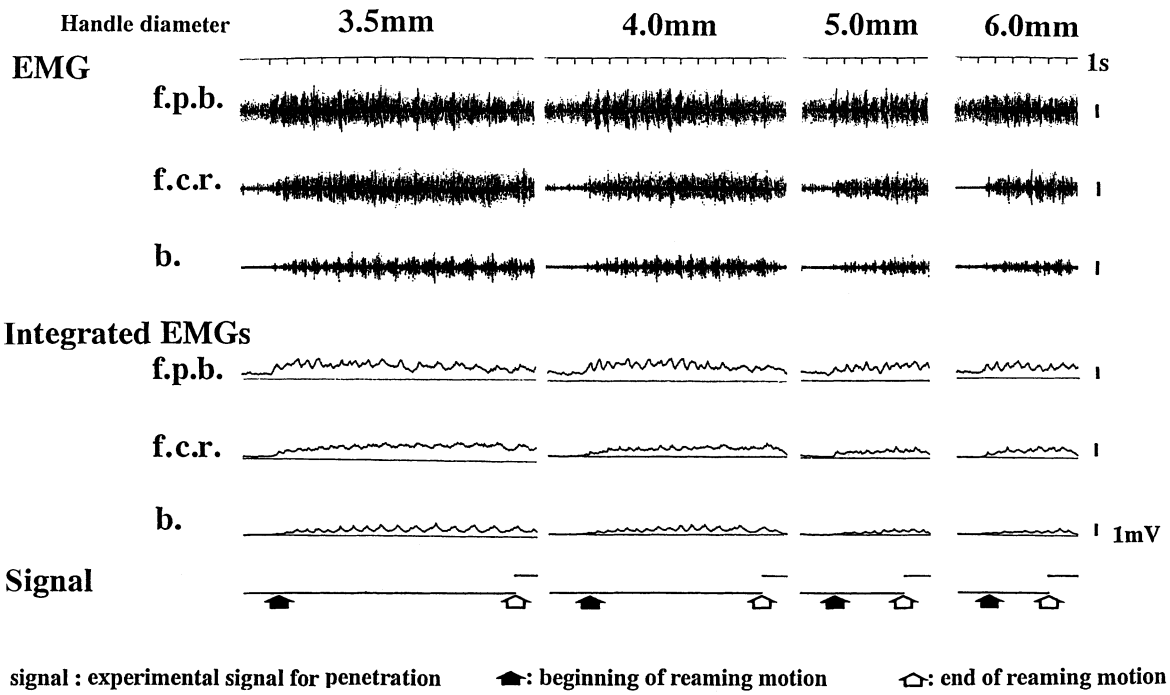

Figure 5 Typical records of EMGs during reaming motion.

Handle diameter(mm)	3.5	4.0	5.0	6.0
Time(s)	12.3±6.9	8.2±7.0	4.5±3.1	4.0±1.9
Integrated EMG area(mV・S)				
flexor pollicis brevis muscle	18.9±9.1	12.2±8.4	6.0±2.9	5.3±2.2
flexor carpi radialis muscle	11.3±10.6	6.2±8.4	2.3±2.3	2.0±1.4
brachioradialis muscle muscle	10.1±9.7	6.2±7.6	2.5±2.2	2.4±2.4
Maximum amplitude of EMG(mV)				
flexor pollicis brevis muscle	2.5±0.2	2.3±0.3	2.1±0.4	2.0±0.4
flexor carpi radialis muscle	1.5±0.5	1.1±0.5	0.9±0.3	1.1±0.6
brachioradialis muscle	1.5±0.4	1.3±0.7	1.1±0.6	1.3±0.9

Table 1 Preparation time, integrated EMG area, and maximum amplitude of EMG for the four handle diameters (mean ± S.D., $n = 7$, $^{*}P < 0.05$, $^{**}P < 0.01$)

In addition, the numbers of broken files and blocked canals, including those used during practice are shown in Table 2. Many blocked canals occurred with 3.5 mm handles (eight files), but the instruments with 6 mm handles did not produce any blocked canals.

Five files fractured; two files had 3.5 mm handles and three files had 6 mm handles.

Discussion

Some studies (Chandler & Bloxham 1990, Treble *et al.* 1993, Chandler *et al.* 1996) have reported the influence on tactile sensitivity and efficiency of handle design or diameter when wearing gloves. However, the optimum handle design remains ambiguous.

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Table 2 The number of fractured files and blocked canals

Handle diameter (mm)	3.5	4.0	5.0	6.0
Fractured file	2 (2/38, 5.3%)	0 (0%)	0 (0%)	3 (3/31, 9.7%)
Blocked canal	8 (8/38, 23.7%)	1 (1/29, 3.4%)	2 (2/30, 6.7%)	0 (0%)

Chandler & Bloxham (1990) reported that a handle of maximum diameter 4.2 mm was always faster than the handle of maximum diameter 4 mm in root canal instrumentation. However, these instrument handles had different designs. In our study different diameters of identical round handles were used in order to investigate specifically the influence of handle diameters.

Four different diameters were fitted on size 45 K-files. Size 45 was selected because the root canal model had a 0.5-mm inner diameter, allowing the tip of the file to be inserted in the canal model and rotated in the canal. The working length of 5 mm was selected from preliminary studies, because it was possible to penetrate that length with size 45 K-type files using only rotational motion, and the time to achieve this could be recorded satisfactorily. The subjects were seven practising dentists with less than 1 year of clinical experience.

In order to determine the most suitable methods and materials required for reducing operator effort during root canal instrumentation, this study focused on the changes in operator effort (using EMG) relating to changes in instrument handle diameter.

Electromyographic amplitudes recorded with bipolar surface electrodes showed the strength of muscle contraction and a pattern for each behaviour. There was a linear relationship between the strength of muscle contraction and the integrated EMG recording (Bigland & Lippold 1954). Integrated EMG area indicated the amount of muscle activity.

Electromyographic recordings of forearm muscles during extension, flexion, abduction, adduction and various pinch motions, etc., have been reported in orthopaedics (Satoh *et al.* 1969). But there have been no reports about operator muscles during dental practice.

Three muscles were selected on palpation during the instrument holding/pinching motion, and to get a clear record of EMG activities the only activity performed was a reaming motion.

Both the integrated EMG area and the maximum amplitude of EMG recorded from the f.p.b. significantly decreased following an increase in handle diameter.

For the f.c.r. and the b there were no significant differences in the maximum amplitude of EMG. A significant difference in the integrated EMG area from the f.c.r. and the b was recorded, but there were variations amongst subjects.

Muscle activities during reaming consisted of synergistic patterns involving many muscles. However, although it is necessary to record the EMG of many other muscles (e.g. hand, forearm and brachialis muscles), it is difficult to perform such recordings because EMG recording of deep stratum muscles requires muscle penetration with fine-wire electrodes and not surface electrodes.

The handle diameters of endodontic instruments on the market are currently in the 3.5~4 mm range. The results of this study suggest that as handle diameters increase from 3.5 mm to 5 mm then instrumentation time becomes shorter and operators' effort reduces.

Conclusion

This experiment suggests that handle size has an effect on reaming time and muscle activity and therefore can influence operator performance during instrumentation.

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